Kaushil Nagrale

+1 (623) 287-8754 | kaushil.nagrale@gmail.com | linkedin.com/in/kaushilnagrale/ | kaushilnagrale.github.io/

EDUCATION

Arizona State University — Tempe, USA
Master's in Data Science, Analytics and Engineering | GPA: 3.90 / 4.00 | $10K NAMU Scholarship — *August 2024 - May 2026*

RV College of Engineering — Bangalore, India
Bachelor's in Mechanical Engineering | GPA: 8.54 / 10.00 — *August 2017 - May 2021*

TECHNICAL SKILLS

Programming Languages: Java, Python, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Security, Spring Data JPA, Spring Batch, Spring AI, FastAPI, React, Zustand, Leaflet

AI/ML: LLMs (GPT, Claude, Llama, Qwen), RAG, LangChain, BM25 Hybrid Search, Cross-Encoder Reranking, Claude Code

Databases and Vector Stores: PostgreSQL, pgvector, Redis, OpenSearch, ChromaDB, FAISS

Cloud and DevOps: AWS, Azure, Azure DevOps, Docker, Apache Kafka, CI/CD, Git

Architecture: Microservices, Event-Driven Architecture, Saga Orchestration, Transactional Outbox, Circuit Breakers, Multi-Tenancy

Security and Testing: OAuth 2.0, JWT, RESTful API Design, JUnit, jqwik Property Testing

PROFESSIONAL EXPERIENCE

Software Engineer (Part Time) — Tempe, USA
Arizona State University — *August 2024 - May 2026*

- Developed barcode-based equipment tracking into a **Spring Boot** and **React** inventory platform using **Claude Code**, reducing processing time by **31%** across a 254,000-square-foot venue hosting **10,000+** annual events.
- Integrated booking and reservation data through **REST APIs** to generate room-setup diagrams document, adding equipment-usage reporting; this enabled event-planning teams to streamline setup workflows and reduce manual effort by estimated **64%**.
- Implemented **role based access control** with **Spring Security** and **JWT**, securing check in, scheduling, and reporting workflows.

Software Engineer — India
Cognizant Technology Solutions — *September 2021 - January 2024*

- Developed **e-commerce web application** for a UK tyre distributor and major retailer, integrating third-party payments and loyalty rewards to seamlessly process orders for **8.6M+** active app users.
- Played a key role in decomposing a monolithic billing service into Spring Boot microservices on Azure Kubernetes Service, using saga orchestration and **Kafka** events with the transactional outbox pattern to improve scalability, resilience, and transaction reliability.
- Cut **p95 checkout latency from 850 ms to 510 ms** by parallelising independent downstream calls, tightening timeouts with **circuit breakers** on external dependencies, and caching hot pricing and catalog lookups in **Redis**.
- Automated delivery through **Azure DevOps** pipelines, **Docker** image builds, and blue green rollouts across 6 services, reducing deployment time from **5 hours to 30 minutes** and enabling rollback by traffic switch rather than redeploy.

PROJECTS

Verdant Live Plant Marketplace

- Built a **multi tenant marketplace** with **saga orchestrated fulfillment**, transit time optimized seller allocation, and outbox based event delivery.
- Implemented a **double entry settlement ledger** verified by **jqwik property testing** against full journal replay.

PulseRoute | Fan Favorite Winner, Kiro Spark Challenge

- Built a **heat safe cycling co-pilot** with a team of 3 in under 24 hours, fusing **Open-Meteo**, **NWS**, **AirNow**, and **Mapbox** data into **real time heat risk routing**.
- Designed the **system architecture and PRD** and built the **onboarding personalization layer**, feeding rider profiles into a safety gate that refuses alerts on missing or stale data, backed by **60+ unit tests** at **100% branch coverage**.

Hybrid RAG Intelligent Ticketing System (Capstone)

- Led a team of 5 to build a **7-stage hybrid RAG pipeline** over an **80K ticket corpus**, combining BM25 and dense retrieval with **reciprocal rank fusion** and **cross-encoder reranking** to reach **97.6% Recall@5** on 500 evaluation queries.
- Designed an **LLM evaluation framework** that cut hallucination from **44% to 2.8%** with **100% out of scope refusal**, served through **FastAPI** and **React**.

CliniShield Caregiver Medication Platform

- Built a drug interaction knowledge base from FDA sources after NLM retired the RxNav interaction API, ingesting **262K labels** via partitioned **Spring Batch** and computing FAERS disproportionality signals over **2M rows**.
- Served a citation enforced **hybrid RAG assistant** across four microservices with three layer **tenant isolation**, gated in CI by an 84 test guardrail suite.

CERTIFICATIONS

Microsoft Certified: Azure Developer Associate (AZ-204) | AWS Certified Developer Associate